Quantification: 543_3
=====================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 13 |
| interior lacuna count | 12 |
| frame-edge lacuna count | 1 |

Interior lacunae only: mean, median and sample SD (n = 12)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2350.2 | 2188.5 | 668.925 | px^2 |
| major_axis_length_px | lacuna shape | 91.5632 | 84.5141 | 33.0603 | px |
| minor_axis_length_px | lacuna shape | 36.5948 | 35.0553 | 10.2044 | px |
| aspect_ratio | lacuna shape | 2.6601 | 2.3867 | 1.276 | unitless |
| eccentricity | lacuna shape | 0.889 | 0.9075 | 0.078 | unitless |
| solidity | lacuna shape | 0.8742 | 0.9204 | 0.1013 | unitless |
| roots_count | headline | 14.25 | 14 | 3.7447 | count |
| ring_length_r30_px | headline | 411.417 | 414 | 106.432 | px |
| ring_length_r60_px | reported | 1076.25 | 1064 | 208.783 | px |
| owned_length_px | ownership-dependent | 3636.52 | 3773.94 | 1147.36 | px |
| edge_count | ownership-dependent | 168.75 | 163 | 61.2983 | count |
| mean_edge_length_px | ownership-dependent | 21.9749 | 22.0258 | 1.7723 | px |
| perimeter_px | normalised | 221.015 | 219.693 | 100.917 | px |
| ring_area_r30_px2 | normalised | 8668.5 | 8576.5 | 2568.76 | px^2 |
| ring_area_r60_px2 | normalised | 21815.7 | 21875 | 4394.01 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9974 | 1 | 0.009 | unitless |
| in_frame_fraction_r60 | normalised | 0.9801 | 1 | 0.058 | unitless |
| ring_density_r30 | normalised | 0.0491524 | 0.0465214 | 0.0115897 | px^-1 |
| ring_density_r60 | normalised | 0.0500524 | 0.0485338 | 0.00781146 | px^-1 |
| roots_per_100px_perimeter | normalised | 8.7387 | 6.2331 | 7.0088 | per 100 px |
| ring_attached_length_r30_px | network v2 | 384.917 | 385.5 | 111.703 | px |
| ring_attached_length_r60_px | network v2 | 970.083 | 989.5 | 235.169 | px |
| ring_length_w_r30_px | network v2 | 467.79 | 465.019 | 121.117 | px |
| ring_length_w_r60_px | network v2 | 1228.26 | 1208.44 | 236.062 | px |
| sholl_crossings_r10 | network v2 | 10.9167 | 10 | 3.9418 | count |
| sholl_crossings_r20 | network v2 | 13.4167 | 14 | 2.6785 | count |
| sholl_crossings_r30 | network v2 | 15.5 | 15.5 | 3.4245 | count |
| loop_count | loops, report only | 19.5 | 20.5 | 11.9507 | count |
| loop_edge_share | loops, report only | 0.4175 | 0.4053 | 0.1426 | share |
| ring_width_mean_r30_px | width | 7.1769 | 7.4567 | 0.9937 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.02406e+06 | px^2 |
| total_skeleton_length_px | 44604 | px |
| canalicular_length_density_per_px | 0.043556 | px^-1 |
| junction_count | 1171 | count |
| junction_density_per_px2 | 0.00114349 | px^-2 |
| lacuna_count | 13 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.269e-05 | px^-2 |
| skeleton_component_count | 142 | count |
| mean_component_length_px | 314.113 | px |
| median_component_length_px | 18.5 | px |
| field_length_density_w_per_px | 0.049656 | px^-1 |
| field_density_without_flagged_per_px | 0.043556 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 211 | px |
| unattached_component_count | 14 | count |
| canalicular_length_density_attached_per_px | 0.043556 | px^-1 |
| field_length_density_with_unattached_per_px | 0.043762 | px^-1 |
| loop_count | 286 | count |
| loop_edge_share | 0.4916 | share |
| edge_count | 2317 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.9003 | px |
| width_median_px | 8 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 44604 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2174 | 80.8669 | 35.2412 | 2.2947 | 0.9 | 0.9518 | 0.4109 | 79.3965 | 270.898 | False |
| 2 | 3447 | 90.3677 | 63.9486 | 1.4131 | 0.7066 | 0.674 | 0.4223 | 197.059 | 916.902 | False |
| 3 | 1883 | 69.2617 | 34.8694 | 1.9863 | 0.864 | 0.9448 | 0.5742 | 250.833 | 340.47 | False |
| 4 | 3536 | 181.867 | 30.1414 | 6.0338 | 0.9862 | 0.7148 | 0.0738 | 327.046 | 42.4075 | False |
| 5 | 2052 | 69.2015 | 39.4228 | 1.7554 | 0.8219 | 0.9112 | 0.0622 | 388.671 | 369.939 | False |
| 6 | 2266 | 88.1613 | 35.5691 | 2.4786 | 0.915 | 0.9021 | 0.1324 | 473.05 | 745.191 | False |
| 7 | 1625 | 80.1248 | 27.6985 | 2.8927 | 0.9383 | 0.8325 | 0.2321 | 545.572 | 877.076 | False |
| 8 | 1683 | 71.0123 | 31.3735 | 2.2634 | 0.8971 | 0.9355 | -0.0518 | 634.68 | 198.315 | False |
| 9 | 2203 | 91.2092 | 31.4053 | 2.9043 | 0.9389 | 0.9295 | 0.0832 | 734.301 | 635.44 | False |
| 10 | 2633 | 93.5592 | 36.2778 | 2.579 | 0.9218 | 0.9454 | 0.0457 | 780.576 | 481.495 | False |
| 11 | 531 | 28.7881 | 24.8252 | 1.1596 | 0.5063 | 0.9332 | 0.5942 | 1012.28 | 184.834 | True |
| 12 | | | | | | | | 458.12 | 481.03 | False |
| 13 | | | | | | | | 857.425 | 285.616 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 15 | 414 | 1151 | 2171.94 | 96 | 22.6243 | 6.8583 |
| 2 | 14 | 603 | 1467 | 4465.98 | 214 | 20.8691 | 7.0779 |
| 3 | 10 | 414 | 1065 | 4562.12 | 207 | 22.0392 | 8.3894 |
| 4 | 21 | 543 | 1063 | 3915.32 | 199 | 19.675 | 7.6996 |
| 5 | 14 | 376 | 970 | 3345.89 | 152 | 22.0124 | 7.7264 |
| 6 | 18 | 425 | 1157 | 2473.82 | 109 | 22.6956 | 8.1921 |

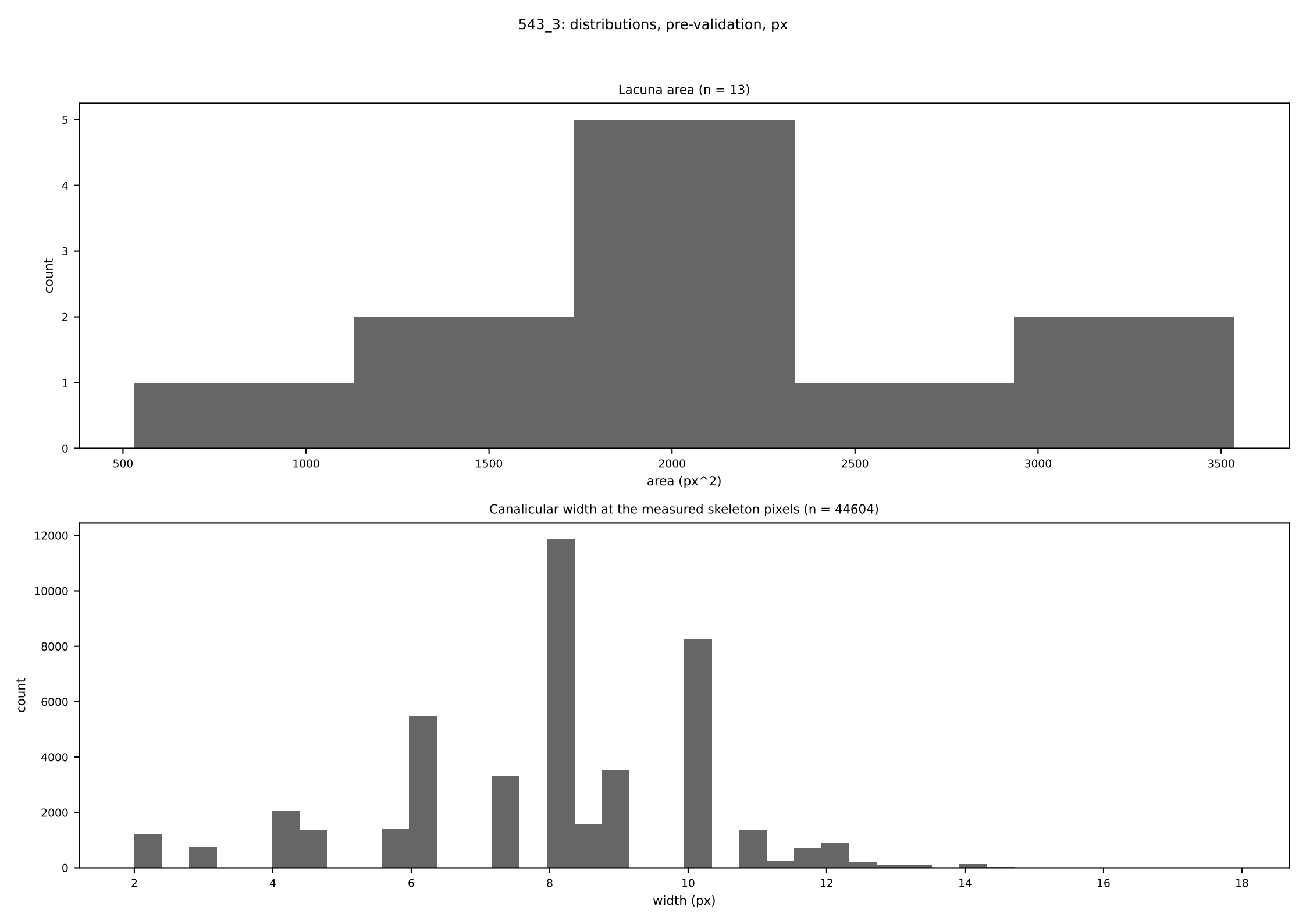
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 7 | 13 | 523 | 1388 | 5730.31 | 293 | 19.5574 | 6.3897 |
| 8 | 16 | 320 | 1027 | 4546.21 | 212 | 21.4444 | 7.6072 |
| 9 | 9 | 432 | 1139 | 4022.11 | 165 | 24.3764 | 7.7086 |
| 10 | 11 | 330 | 905 | 1757.89 | 69 | 25.4767 | 7.3063 |
| 11 | 12 | 193 | 577 | 1477.47 | 75 | 19.6996 | 6.3485 |
| 12 | 11 | 333 | 855 | 3632.55 | 161 | 22.5624 | 4.7196 |
| 13 | 19 | 224 | 728 | 3014.08 | 148 | 20.3654 | 6.4471 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 210.066 | 8595 | 21715 | 1 | 0.9616 | 0.0481675 | 0.0530048 | 7.1406 |
| 2 | 363.446 | 12097 | 29071 | 1 | 1 | 0.0498471 | 0.0504627 | 3.852 |
| 3 | 184.794 | 7941 | 19515 | 1 | 1 | 0.0521345 | 0.0545734 | 5.4114 |
| 4 | 415.99 | 13536 | 26753 | 0.9687 | 0.7992 | 0.0401152 | 0.0397339 | 5.0482 |
| 5 | 205.965 | 8333 | 20013 | 1 | 1 | 0.0451218 | 0.0484685 | 6.7973 |
| 6 | 241.907 | 9380 | 23807 | 1 | 1 | 0.0453092 | 0.0485992 | 7.4409 |
| 7 | 229.321 | 8558 | 22035 | 1 | 1 | 0.0611124 | 0.0629907 | 5.6689 |
| 8 | 191.965 | 7943 | 21525 | 1 | 1 | 0.040287 | 0.047712 | 8.3348 |
| 9 | 238.108 | 9212 | 24077 | 1 | 1 | 0.0468954 | 0.0473066 | 3.7798 |
| 10 | 242.936 | 9386 | 24420 | 1 | 1 | 0.0351588 | 0.0370598 | 4.5279 |
| 11 | 95.3909 | 3354 | 9525 | 0.6102 | 0.5725 | 0.0575432 | 0.0605774 | 12.5798 |
| 12 | 47.4558 | 4187 | 13513 | 1 | 1 | 0.0795319 | 0.0632724 | 23.1794 |
| 13 | 80.2254 | 4854 | 15344 | 1 | 1 | 0.0461475 | 0.0474453 | 23.6833 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 379 | 981 | 470.718 | 1335.28 | 9 | 15 | 18 |
| 2 | 578 | 1374 | 696.955 | 1672.82 | 19 | 16 | 19 |
| 3 | 392 | 1049 | 459.321 | 1211.83 | 10 | 13 | 18 |
| 4 | 543 | 1010 | 608.174 | 1205.05 | 17 | 16 | 10 |
| 5 | 365 | 903 | 420.635 | 1116.52 | 10 | 15 | 14 |
| 6 | 425 | 998 | 477.777 | 1311.95 | 11 | 14 | 14 |
| 7 | 466 | 1321 | 602.801 | 1582.25 | 12 | 16 | 22 |
| 8 | 289 | 936 | 367.392 | 1169.28 | 9 | 10 | 15 |
| 9 | 413 | 1051 | 489.676 | 1281.93 | 13 | 14 | 16 |
| 10 | 257 | 648 | 374.291 | 1022.26 | 9 | 14 | 13 |
| 11 | 153 | 333 | 220.51 | 657.458 | 4 | 5 | 9 |
| 12 | 314 | 817 | 382.877 | 996.134 | 7 | 10 | 16 |
| 13 | 198 | 553 | 262.865 | 833.796 | 5 | 8 | 11 |


543_3: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 44604)
count
width (px)